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Project Title:	HZE Radiation: Modulation Of Genetic Effects By RNA Interference Of NHEJ		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
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COI Name (Institution):	Bedford, Joel (Colorado State University) Liber, Howard (Colorado State University)		
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Task Description:

For extended missions in space, astronauts will be exposed to a radiation environment unlike any on the earth's surface. Galactic cosmic rays include a particularly worrisome component of energetic iron nuclei which are high atomic number, high energy (HZE) charged particles with linear energy transfer (LET) similar to alpha particles but with very different track structures. While the relative biological effectiveness (RBE) of a particular radiation roughly depends on the LET, the very different track structures for these two radiations may well modulate this RBE even if they have identical average LETs. Further, this "radiation quality" for the exposures actually relevant for astronauts will be modified by interactions with materials in the spacecraft, including their own bodies.
Basic to virtually all relevant biological effects of ionizing radiation is the underlying damage produced in DNA, and the subsequent cellular processing of such damage. The damage can be qualitatively different for different kinds of radiations, and the genetics of the biological systems exposed can greatly affect damage processing and ultimate outcome—the biological effect of concern. The accurate repair of DNA double-strand breaks (DSB) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosome rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. It is important, therefore, to understand cellular responses to DNA damage and to elucidate the mechanisms and phenotypic consequences of DSB misrepair.

Our general aim in this proposal is to examine connections between cellular radiosensitivity phenotypes relevant for carcinogenesis, namely chromosome aberration formation, mutagenesis and telomere fusions, and deficiencies in various components of the non-homologous end joining system, using RNA interference to systematically knockdown gene products known to be involved, either singly or in combination.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

This project utilizes the powerful technique of RNA interference (RNAi) to deplete levels of various components of the non-homologous end joining (NHEJ) double-strand break (DSB) repair system, and to study the effects on HZE radiation-induced chromosomal rearrangements and gene mutation.
The accurate repair of DNA double-strand breaks (DSBs) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosomal rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. We have focused on the consequent phenotypic effects of faulty repair by examining connections between cellular radiosensitivity phenotypes relevant for carcinogenesis after exposure to IR, and deficiencies in various components of the non-homologous end joining (NHEJ) system. In the past year, we produced deficiencies of Artemis and Apollo using RNAi in a human lymphoblastoid cell line. Exposure of cells exhibiting reduced protein expression to either gamma-rays or 1GeV/nu Fe particles demonstrated differential effects on telomere dysfunction and mutation frequency, as well as differential effects between radiation qualities.

Regarded as a “guardian” of DNA ends, Artemis is a NHEJ accessory factor that is mutated in RS-SCID individuals. Artemis belongs to the SNM-1 family of proteins, possessing a metallo-ß-lactamase domain and a ß-CASP region, suggestive of interaction with nucleic acids. Upon DSB induction and hypophosphorylation by ATM, Artemis exhibits intrinsic 5’-3’ single-strand DNA exonuclease activity. DNA-PKcs autophosphorylation at the Thr-2609 cluster is required for activation of Artemis endonuclease activity. Interestingly, we have shown that DNA-PKcs autophosphorylation at this site is required for effective mammalian telomeric end-capping function. It is believed that autophosphorylation-mediated conformational changes in DNA-PKcs regulate Artemis access to DNA ends and facilitate necessary end-processing, a critical event at both proper DSBs and natural chromosomal termini (telomeres). Artemis appears to be involved in mediating the processing of otherwise non-ligatable IR-induced DSBs; a subset of ~10-15%. We hypothesized, therefore, that Artemis activity and end-processing may be especially important following HZE exposure and the induction of complex types of damage.

We initiated studies with inhibition of Artemis in 2006, either alone or in conjunction with DNA-PKcs. The results are quite intriguing. Knockdown of Artemis alone had no effect on spontaneous mutagenesis or on radiation-induced mutagenesis, for either HZE or gamma-rays. However, knockdown of Artemis appeared to prevent the enhancement of mutagenesis that is mediated by inhibition of DNA-PKcs. This suggests that the presence of Artemis is necessary for increased mutagenesis to occur in the absence of DNA-PKcs.

We also found evidence of HZE-induced telomere uncapping/fusion following knockdown of Artemis; this effect was more pronounced than after DNA-PKcs inhibition, or in fact for the two together. These results support Artemis acting in concert with DNA-PKcs to process complex break ends, thus being especially important following HZE exposure. These results are also consistent with our observed increased mutagenesis following HZE. We also noted evidence of telomere duplication (inappropriate homologous recombination) with knockdown of Artemis.

We speculate that without DNA-PKcs, Artemis is activated by ATM in a backup/slow component of NHEJ. Without Artemis, ATM initiates cell cycle checkpoint arrest, so there is persistent residual damage. Without either DNA-PKcs or Artemis, levels return to mock.

Apollo is a nuclease related to Artemis. It seems to be required for DNA interstrand cross-link repair, and in fact it appears to interact with FANCD2, also involved in that repair pathway. In addition, Apollo may be involved in a more generalized response to DNA damage, in that it may stimulate the activation of ATM, and it seems to interact with the MRN complex, which is a focal point for several modes of repair.

The Apollo nuclease is also implicated in the protection of telomeres; it interacts with TRF2 in S-phase, and actually is an attractive candidate for the processing of telomere ends to generate 3’ single strand overhangs, a critical event and integral part of proper telomere formation.

Knockdown of Apollo produced an increase in radiation mutagenesis, both for ?-ray exposures and for 1 GeV/n Fe particles. The increases were of the same order as those caused by inhibition of DNA-PKcs, and furthermore there were no interactions, synergistic or antagonistic, of Apollo knockdown and DNA-PKcs inhibition. These results suggest that Apollo may be involved in the same pathway as DNA-PKcs, namely repair of DSB by NHEJ. We also examined telomere instability following siRNA knockdown of Apollo. We found telomere uncapping with knockdown of Apollo

	that was similar to that seen with inhibition of DNA-PKcs and similar between radiation types. These results suggest possible Apollo involvement in repair of IR-induced damage, as well as support a telomeric role for Apollo. Our continuing experiments will lead to valuable information regarding CPR question 10.12, “What are the effects of inter-individual diversity?” We originally hypothesized that NHEJ repair components are major genetic determinants controlling individual susceptibility to radiation-induced cancer, as well as heritable effects of exposure. Therefore, naturally occurring deficiencies of these proteins within the population will put these individuals at increased risk. Our RNAi approach of experimentally mimicking highly relevant human deficiencies continues to support the concept that inherited or acquired mutations in NHEJ and in telomere protection factors represent important risk factors for developing cancer. Our work also impacts on CPR question 10.07, “What are the cancer risks in humans from spaceflight?” Our characterization of telomere dysfunction, chromosome aberration formation and gene mutation will serve as informative surrogate endpoints for predicting cancer risks. Furthermore, we are learning more about the complexities of the DNA repair processes and the inter-relationships between telomeres, repair, and aging.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Peer-reviewed Journals	Bailey SM. "2007 Michael Fry Research Award Lecture: Telomeres and double-strand breaks - all's well that “ends“ well." Radiat Res. 2008 Jan;169(1):1-7. PMID: 18159961 , Jan-2008
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Awards	Bailey SM. "2007 Michael Fry Research Award Lecture: Telomeres and double-strand breaks - all's well that “ends“ well.“ Radiat Res. 2008 Jan;169(1):1-7. PMID: 18159961" Jan-2008